CPE 381: Fundamentals of Signals and Systems for Computer Engineers

01 Introduction and Preliminaries

Fall 2026

Rahul Bhadani

Outline

About Me

Rahul Bhadani
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Research Interests

Cyber-physical Systems, Robotics, Intelligent Transportation, Connected-and-Autonomous Driving, Applied machine learning, Quantum Information Science

Course Logistics

Lecture:
M/W 11:20 AM - 12:40 PM

Location:
ENG 240

Prerequisites:

- EE 213 - Electrical Circuit Analysis I
- MA 238 – Applied Differential Equations

Office Hours:

- Monday/Wednesday: 12:15 PM - 4 PM

Instructor Email: rahul.bhadani@uah.edu

Textbooks

Required: *Signals and Systems using Matlab*. Luis F. Chaparro.
Elsevier, 3rd Edition, 2019.
ISBN: 978-0-12-814204-2, eBook ISBN: 9780128142059

Suggested Reading: *Linear Systems and Signals*, B. P. Lathi, Roger Green, Oxford University Press, 2017, 3rd Edition

Other Notable Textbooks:

- Schaum's Outline of Signals and Systems
- *Signals and Systems*. Haykin, Simon, and Barry Van Veen. John Wiley & Sons, 2007.
- Asadi, Farzin. Signals and Systems with MATLAB and Simulink. Springer, Dec 2023. ISBN: 9783031456220, 303145622X

Grading

Attendance	1%
Handwritten Course Notes	5%
Quizzes	5%
Classwork	14%
Homework	15%
Mid-term Exam 1	15%
Mid-term Exam 2	15%
Final Exam	30%
Total	**100%**

Grading Scale	
Percentage	**Grade**
90% - 100%	A
75% - 89%	B
60% - 74%	C
45% - 59%	D
0% - 44%	F

Homework Policy

- Each late submission will be penalized by 10% per day for up to 5 days maximum, thereafter, if later, one will receive 0 credit.
- Solution to homework will be posted at least 5 days after the due date.

Classwork

- Each lecture will be followed by in-class problem-solving that students will turn in the same lecture day usually, or the next lecture day, depending on the extent of the work. If you miss the lecture day (either the day it is handed to you, or the day you need to turn in), you will not receive any credit.
- There will be intermediate small tests to assess your skills based on lectures and the classwork. This portion will count towards your classwork credits.

Handwritten Course Notes

- You are required to maintain a dedicated course notebook for the course where you will duly record your notes.
- Handwritten course notes will carry 5% towards the final grade.
- Use proper handwriting.

Attendance Policy

- Must attend all lectures.
- Attendance will be taken at the middle of the class.
- Two unexcused absences permitted.
- No option to make up for classwork.

Exam Schedule

Exam Dates		
Exam	**Date**	**Time**
Mid-Term 1	Monday, September 28, 2026	11:20 AM to 12:40 PM
Mid-Term 2	Monday, November 09, 2026	11:20 AM to 12:40 PM
Final Exam	Friday, December 11, 2026	11:30 AM to 2:00 PM

Tentatative Topics

- Introduction, Mathematical Preliminaries
- Continuous and Discrete Signals
- Linear-time Invariant (LTI) Systems
- Laplace Transform
- Fourier Series for Frequency Analysis
- Fourier Transform
- Sampling Theory
- Discrete-Time Signals and Systems
- Z-transform
- Discrete Fourier Analysis

Refer to the syllabus for the detailed information on the syllabus.

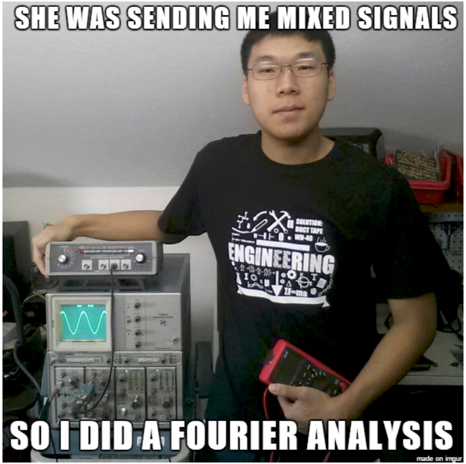
In-Class Activity

Introduce Yourself

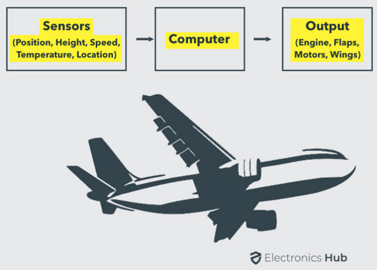
- Why Computer Engineering?
- What are you hoping to learn from this course?
- What is something interesting you did during summer?

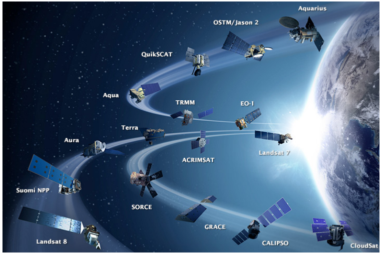
Motivation

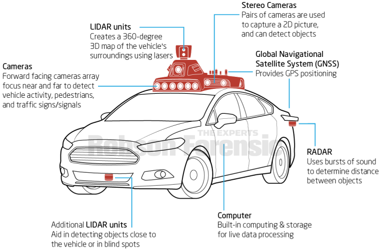
Why Study Signals and Systems



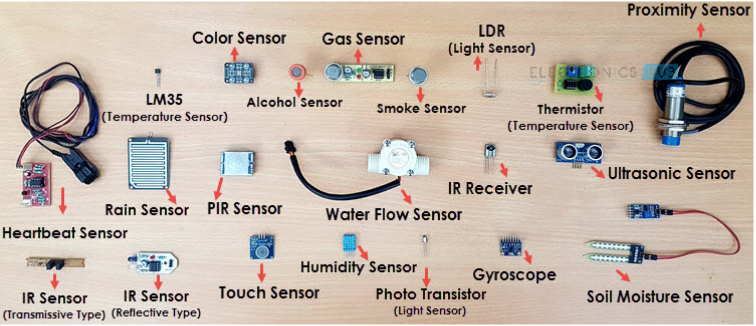
We are in digital era



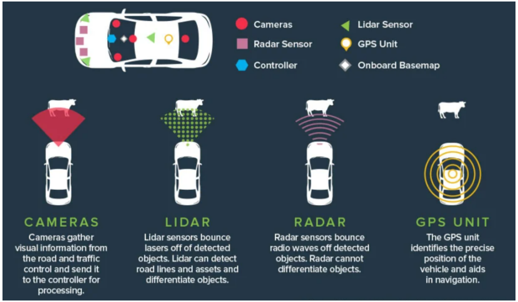




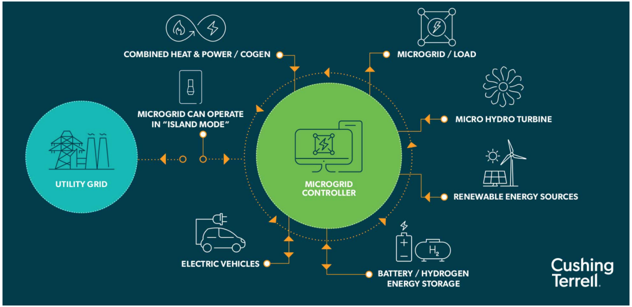
Sensors and Mobile Devices



Cyber-physical Systems



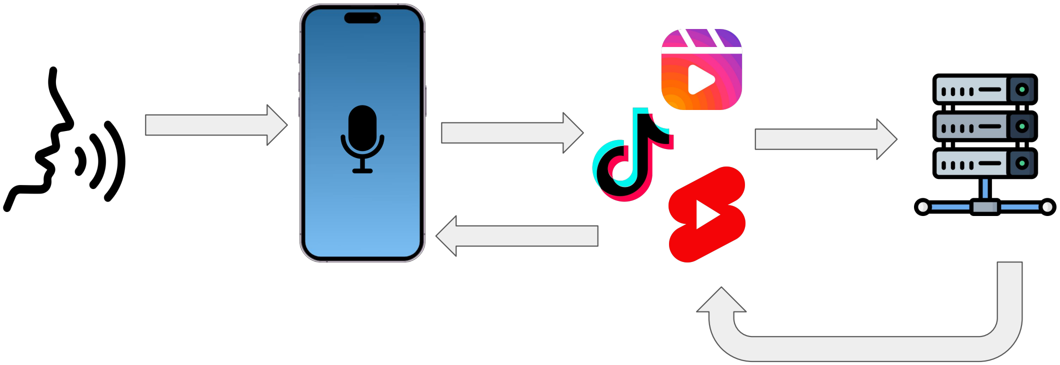
Connected and Autonomous Vehicles



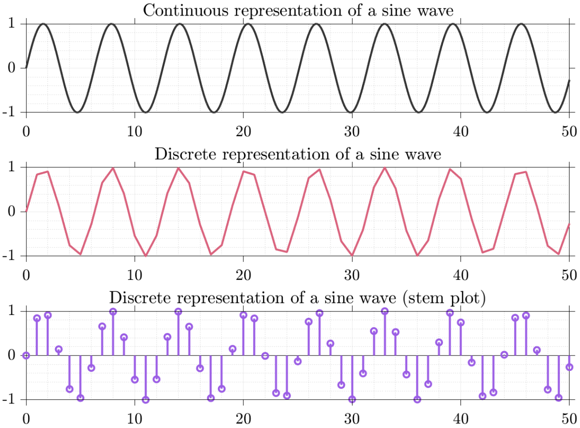
Microgrid

Signals in Nature

Everything in nature is analog and continuous. We use hardware/software interfaces to digitize them, extract information, make inferences, and send them back to the user.



Sampling continuous time signals



$x[n] = x(nT_s)$, T_s = sample time.

Code for the figure: `https://github.com/rahulbhadani/CPE381_FA26/blob/main/Code/Ch01_sampled_sine_wave.m`

Inherent Discrete Time Signals

Nature may be continuous but human activities naturally lead to some discrete time signals.

Examples:

- Stock market closing data
- Adaptive cruise control states
- Thermostat setpoints

Course at A Glance

System Thinking

time	data1	data2
0.01	10	
0.02	11.1	
0.03	11.3	

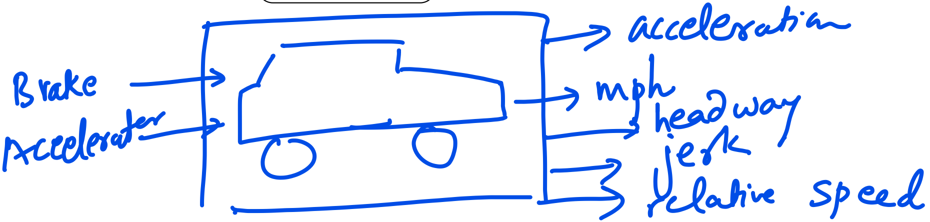
Timeseries

features.

System is an abstraction.

Think of a device or a process as a system – something that transforms one signal into another.

$$x(t) \longrightarrow \boxed{\mathcal{S}} \longrightarrow y(t) = \mathcal{S}[x(t)]$$



Linear Time-Invariant Systems

Linearity of a System $\mathcal{S}$:

$$\mathcal{S}[\overbrace{a \cdot x(t) + b \cdot y(t)}^{z(t)}] = \mathcal{S}[a \cdot x(t)] + \mathcal{S}[b \cdot y(t)] = a \cdot \mathcal{S}[x(t)] + b \cdot \mathcal{S}[y(t)]$$

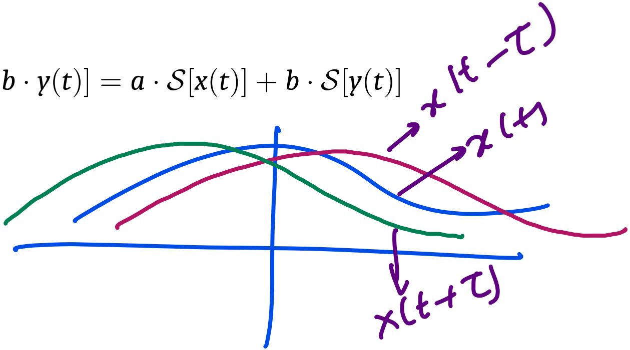
Time-invariance of a System $\mathcal{S}$:

If

$$x(t) \Rightarrow y(t) = \mathcal{S}[x(t)],$$

then, the time-invariance says:

$$x(t \mp \tau) \Rightarrow y(t \mp \tau) = \mathcal{S}[x(t \mp \tau)].$$

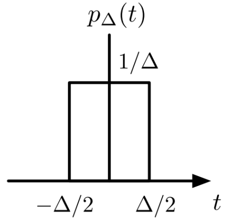


A system satisfying both are Linear Time-Invariant Systems (LTI) Systems.

Rectangular pulse and Unit impulse

- A rectangular pulse of duration Δ and unit area:

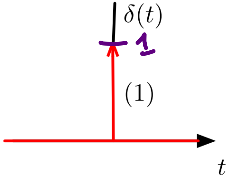
$$p_\Delta(t) = \begin{cases} \dfrac{1}{\Delta}, & -\Delta/2 \le t \le \Delta/2 \\ 0, & \text{otherwise} \end{cases}$$



- Unit Impulse:

$$\delta(t) = \lim_{\Delta \to 0} p_\Delta(t)$$

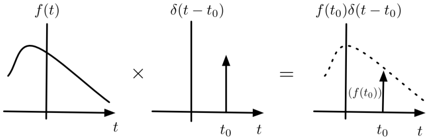
Dirac delta

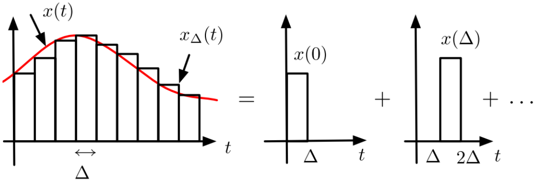


All Signals can be Written Using δ Function

If we do integration in terms of variable τ, we get a generic representation of signals in terms of impulse and shifted impulse.

$$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$$





Approximation of $x(t)$:

$$x_\Delta(t) = \sum_{-\infty}^{\infty} x_\Delta(t-k\Delta) = \sum_{-\infty}^{\infty} x(k\Delta)p_\Delta(t-k\Delta)\Delta$$

In the limit as $\Delta \to 0$ these pulses become impulses, separated by an infinitesimal value:
$\lim_{\Delta\to 0} x_\Delta(t) \to x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$

Representation of Dynamic Systems

Dynamic systems can be represented by ordinary differential equations (ODE) with constant coefficients:

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \cdots + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_m \frac{d^m x(t)}{dt^m} + b_{m-1} \frac{d^{m-1} x(t)}{dt^{m-1}} + \cdots + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

with n initial conditions, $x(t) = 0$ for $t < 0$.

This equation represents a linear ordinary differential equation with constant coefficients, where $y(t)$ is the output, $x(t)$ is the input, and a_i and b_i are the constant coefficients.

Impulse Response

Output when a system is presented with a unit-impulse signal is called **response to an impulse signal**, or in short **impulse response**.

What would we get if we pass $x(t)$ through an LTI system $\mathcal{S}$?

$\delta(t) \rightarrow \boxed{S} \rightarrow h(t)$

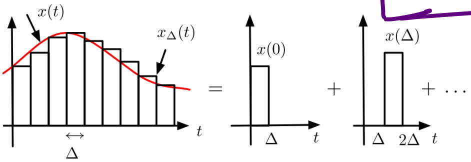
We get

Convolutional integral

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

$x(t) \rightarrow \boxed{S} \rightarrow ?$

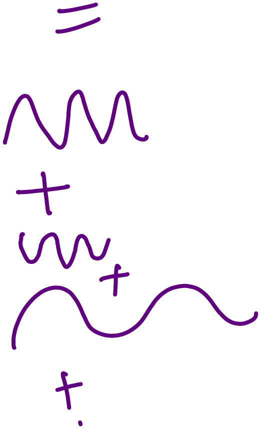
where $h(t)$ is the impulse response. Output is the superposition of the responses of each term in the figure.



Periodic Function’s Fourier Series

If we consider periodic functions with period T_0, such that $x(t) = x(t + T_0)$, then

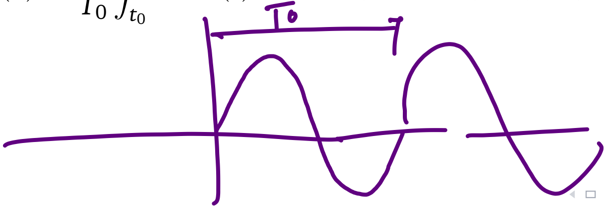
$$x(t) = \sum_{-\infty}^{\infty} \underbrace{X(k)}_{\text{Coefficient}} \overbrace{e^{jk\Omega_0 t}}^{\text{Harmonics}}, \quad \Omega_0 = 2\pi/T_0$$



is called **Fourier Series**.

$$X(k) = \frac{1}{T_0} \int_{t_0}^{t_0+T_0} x(t) e^{-jk\Omega_0 t} dt, \quad k = 0, \pm 1, \pm 2, \ldots$$

is **Fourier coefficients**.



Frequency Response $H(jk\Omega_0)$

$H(k\Omega_0)$ or $H(\Omega_0)$

Frequency Response

$$H(jk\Omega_0) = \int_{-\infty}^{\infty} h(\tau) e^{-jk\Omega_0 \tau} d\tau$$

The frequency response completely characterizes how an LTI system processes periodic signals!

Response of LTI Systems to Periodic Signal

$$x(t) = \sum_k X_k e^{jk\Omega_0 t}, \quad \Omega_0 = \frac{2\pi}{T_0} \tag{1}$$

The output in the steady state, if the impulse response is $h(t)$:

$$y(t) = \sum_{k=-\infty}^{\infty} [X_k H(jk\Omega_0)] e^{jk\Omega_0 t} \tag{2}$$

Fourier Coefficients of $y(t)$ is $Y_k = X_k H_k(jk\Omega_0)$.

Aperiodic Signal

$$f = \frac{1}{T_0}$$

- Aperiodic signal can be thought of as a periodic signal with infinite fundamental frequency.
- In practice, there are no periodic signals, but they are nice mathematical tools.

$$x(t) = \lim_{T_0 \to \infty} \tilde{x}(t)$$

Fourier Transform

In the limiting case, $T_0 \to \infty$,

Fourier Transform

Variable

$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

Inverse Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$

Fourier Transform of Periodic Signals

If the function is periodic, we know that it can be represented as a Fourier series, i.e.

$$x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j\Omega_k t}$$

where

$$X_k = \frac{1}{T}\int_{-\infty}^{\infty} x(t) e^{-j\Omega_k t} dt$$

From the linearity and frequency-shifting property, we can write

$$X(\Omega) = \sum_k \mathcal{F}[X_k e^{j\Omega_k t}] = \sum_k 2\pi X_k \delta(\Omega - \Omega_k)$$

$\Omega_k = k\Omega_0$.

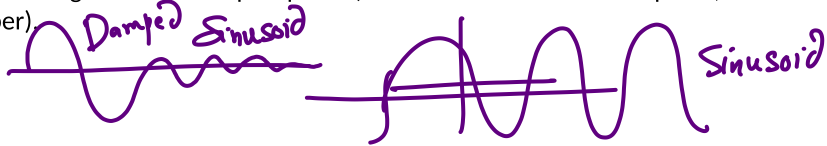
Two-sided or Bilateral Laplace Transform

Definition

$$F(s) = \mathcal{L}[f(t)] = \int_{-\infty}^{\infty} f(t)e^{-st}dt, \quad s \in \mathrm{ROC} \tag{3}$$

ROC = Region of Convergence, the region in the complex plane (henceforth referred to as s plane, as $s = \sigma + j\Omega$ is a complex number).

- σ : damping factor;
- Ω : frequency in $\mathtt{rad/s}$.



We see that in Fourier Transform, a signal is represented as a sum of (weighted) sinusoids. However, in Laplace transform, it is a sum of (weighted) damped sinusoids.

Inverse Laplace Transform

Definition

$$f(t) = \mathcal{L}^{-1}[F(s)] = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st}ds, \quad \sigma \in \mathrm{ROC} \tag{4}$$

The relationship between a function and its Laplace transform can be written as:

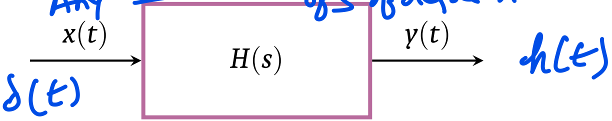
$$f(t) \iff F(s) \tag{5}$$

Transfer Function

For a given system $\mathcal{S}$, if $x(t) = \delta(t)$, then $H(s)$ is called the Transfer Function and characterizes the system in s-domain (or frequency domain).

$$H(s) = \frac{2s^2+4}{3s+5} = \frac{\text{Any Polynomial of } s \text{ of degree } m}{\text{Any} \ldots \text{ of } s \text{ of degree } n}$$

$$\mathcal{L}[h(t)] = H(s)$$



Transfer Function

$$H(s) = \frac{\mathcal{L}[\text{output}]}{\mathcal{L}[\text{input}]} = \frac{\mathcal{L}[y(t)]}{\mathcal{L}[x(t)]} = \frac{Y(s)}{X(s)}$$

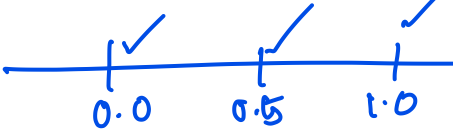
This function is called **transfer function** because it transfers the Laplace transform of the input to the output. Just as with the Laplace transform of signals, $H(s)$ characterizes an LTI system by means of its poles and zeros. Thus it becomes a very important tool in the analysis and synthesis of systems.

Modeling Sampled Signal using Impulses

$T_s = 0.5$

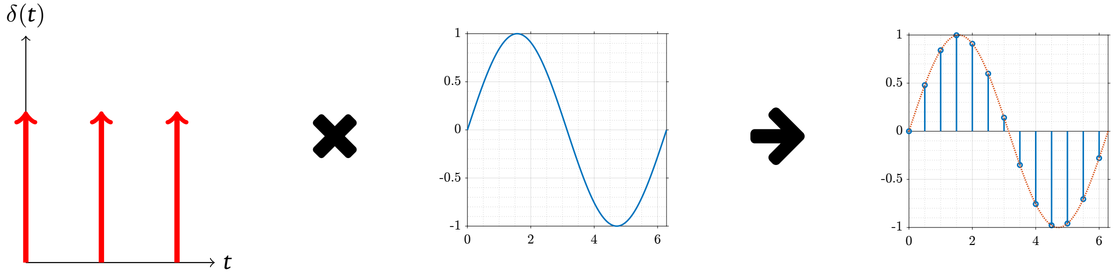
If $\Delta \to 0$, we have a train of impulses. In the limiting conditions, we define the impulse **sampling function** as

$$\delta_{T_s}(t) = \sum_n \delta(t - nT_s)$$



Thus, the sampled signal is:

$$x_s(t) = x(t)\delta_{T_s}(t)$$



Sampled Signal in Continuous-Time and Discrete-Time

$\Omega_s = \frac{1}{T_s}$

Fundamental Frequency

Two ways to imagine the sampled signal $x_s(t)$:

Continuous-Time Version

$$x_s(t) = x(t)\delta_{T_s}(t) = \frac{1}{T_s}\sum_{k=-\infty}^{\infty} x(t)e^{jk\Omega_s t}$$

$$X_s(\Omega) = \frac{1}{T_s}\sum_{k=-\infty}^{\infty} X(\Omega - k\Omega_s)$$

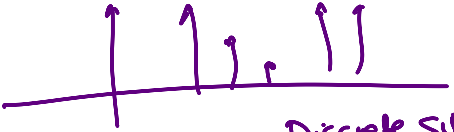
Discrete-time Version

$$x_s(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s)$$

$x[n] = x(nT_s)$

$$X_s(\Omega) = \sum_{n=-\infty}^{\infty} x(nT_s)e^{-j\Omega T_s n}$$

Z-transform from Laplace Transform



Discrete signal

The discrete equivalent of Laplace Transform that follows:

- If the sampled signal is: $x(t) = \sum x(nT_s)\delta(t - nT_s)$
  its Laplace transform is : $X(s) = \sum_n x(nT_s)\mathcal{L}[\delta(t - nT_s)] = \sum_n x(nT_s) = e^{-nsT_s}$
- By letting $z = e^{sT_s}$,

$$\mathcal{Z}[x(nT_s)] = \mathcal{Z}[x[n]] = \mathcal{L}[x_s(t)]|_{z=e^{sT_s}} = \sum_n x(nT_s)z^{-n} = \sum_n x[n]z^{-n}$$

Z-transform

which is called the Z-transform of the sampled signal.

$\{3.2, 4.1, 3.9\}$

$n=0 \quad n=1 \quad n=2$

$X(z) = 3.2 \cdot z^{-0} + 4.1 \times z^{-1} + 3.9 \times z^{-2}$

I	D
0.0	3.2
0.5	4.1
1.0	3.9
⋮	

Two sided Z-transform

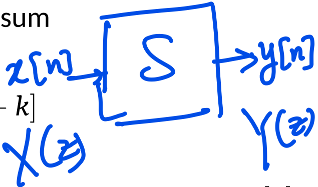
Given a discrete-time signal $x[n]$, $-\infty < n < \infty$, its two-sided Z-transform is given by

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

defined in a region of convergence (ROC) on the Z-plane.

Transfer Function in Z-domain

The output $y[n]$ of a causal LTI system is calculated using the convolution sum



$$y[n] = [x * h][n] = \sum_{k=0}^{n} x[k]h[n-k] = \sum_{k=0}^{n} h[k]x[n-k]$$

where $x[n]$ is a causal input and $h[n]$ is the impulse response of the system. Z-transform of the $y[n]$ is the product

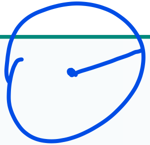
$$Y(z) = \mathcal{Z}\{[x * h][n]\} = \mathcal{Z}\{x[n]\}\mathcal{Z}\{h[n]\} = X(z)H(z)$$

and thus the transfer function of a discrete system is defined as

$$H(z) = \frac{Y(z)}{X(z)}$$

The Discrete-Time Fourier Transform (DTFT)

For a discrete-time signal $x[n]$, the DTFT is



$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}, \quad -\pi \leq \omega \leq \pi \tag{6}$$

and the inverse DTFT is

$$x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega \tag{7}$$

Discrete Fourier Transform (DFT)

- In DTFT, the frequency ω varies from $-\pi$ to π, so computing $X(e^{j\omega})$ will need to be for uncountable frequencies, not possible for computers.
- The inverse DTFT requires integration which cannot be exactly implemented in computers.

So we consider Discrete Fourier Transform (DFT) which is computed at discrete frequencies and its inverse also doesn't require integration.

We can write the DFT of a periodic signal $\tilde{x}[n]$ as

$$X[k] = N\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n]e^{-j\omega_0 nk} = \tilde{x}[n]e^{-j2\pi nk/N}, \quad 0 \leq k \leq N-1, \omega_0 = 2\pi/N$$

and the inverse DFT is

$$\tilde{x}[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]e^{j2\pi nk/N}, \quad 0 \leq n \leq N-1$$

DFT of Aperiodic Signal

- Consider a discrete signal $y[n]$ of finite length N.
- Choose an integer $L \geq N$, the length of the DFT, to be the fundamental period of a periodic extension $\tilde{y}[n]$ having $y[n]$ as a period with padded zeros if necessary.

So, DFT of $\tilde{y}[n]$ is

$$\tilde{y}[n] = \frac{1}{L}\sum_{k=0}^{L-1} \tilde{Y}[k]e^{j2\pi nk/L} \quad 0 \leq n \leq L-1$$

and inverse DFT is

$$\tilde{Y}[k] = \sum_{n=0}^{L-1} y[n]e^{-j2\pi nk/L} \quad 0 \leq k \leq L-1$$

The Fast Fourier Transform (FFT)

- FFT efficiently computes the **DFT**. It is not a new transform.
- Utilizes a divide-and-conquer strategy (e.g., Cooley-Tukey algorithm).
- Drastically reduces computational complexity:

$$O(N^2) \quad \longrightarrow \quad O(N \log N)$$

Makes modern digital signal processing practical.

$$X[k] = \sum_{n=0}^{N-1} x[n]\, W_N^{kn}, \quad \text{where } W_N = e^{-j\frac{2\pi}{N}}$$

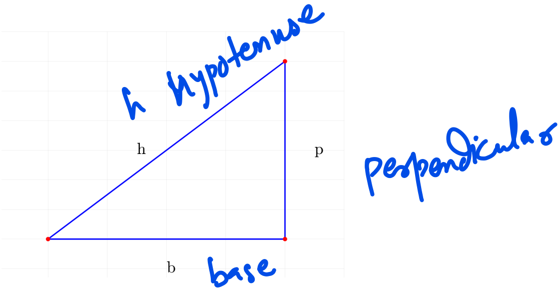
$$X[k] = \underbrace{E[k]}_{\text{even indices}} + W_N^k \underbrace{D[k]}_{\text{odd indices}}$$

Mathematical Preliminaries

Trigonometry

- **Sine, Cosine, Tangent:**
  - $\sin(\theta) = \frac{\mathsf{p}}{\mathsf{h}}$
  - $\cos(\theta) = \frac{\mathsf{b}}{\mathsf{h}}$
  - $\tan(\theta) = \frac{\mathsf{p}}{\mathsf{b}}$
- **Pythagorean Identity:**

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

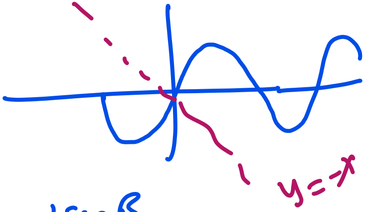


- **Unit Circle:**

$h = 1$

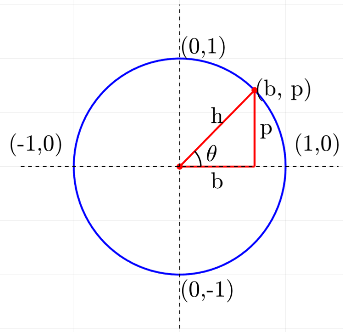
$h^2 = p^2 + b^2$

Trigonometric Identities

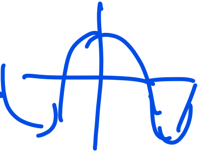


What is the value of h based on the diagram?

$h = 1$



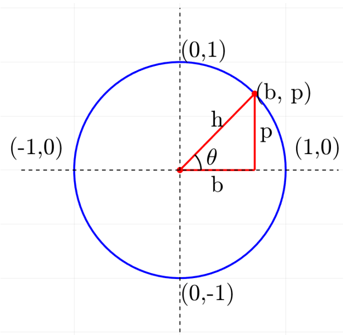
- $\cos\theta =$ $b = \frac{b}{h} = \frac{b}{1} = b$
- $\sin\theta =$ p
- $\sin(\alpha+\beta) =$ $\sin\alpha\cos\beta + \cos\alpha\sin\beta$
- $\cos(\alpha+\beta) =$ $\cos\alpha\cos\beta - \sin\alpha\sin\beta$
- $\sin^2(\theta) =$ $\frac{1-\cos 2\theta}{2}$ in terms of $\cos(2\theta)$
- $\cos^2(\theta) =$ $\frac{1+\cos 2\theta}{2}$ in terms of $\cos(2\theta)$
- $\sin 2\theta =$ $2\sin\theta\cos\theta$ in terms of θ
- $\cos 2\theta =$ $1 - 2\sin^2\theta$ in terms of θ
- $\sin(-\theta) =$ $\sin\theta$
- $\cos(-\theta) =$ $\cos\theta$



Trigonometric Identities

What is the value of h based on the diagram?

$h = 1$

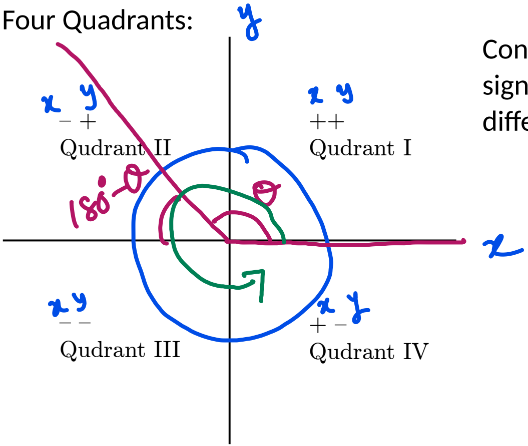


- $\tan(\alpha + \beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$
- $\tan(\alpha - \beta) = \dfrac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$
- $\sin 30^\circ = 1/2$
- $\sin 45^\circ = 1/\sqrt{2}$
- $\sin 60^\circ = \sqrt{3}/2$
- $\cos 30^\circ = \sqrt{3}/2$
- $\cos 45^\circ = 1/\sqrt{2}$
- $\cos 60^\circ = 1/2$
- $\tan 30^\circ = 1/\sqrt{3} = \dfrac{\sin 30}{\cos 30} = \dfrac{1}{\cancel{2}} \cdot \dfrac{\cancel{2}}{\sqrt{3}} = \dfrac{1}{\sqrt{3}}$
- $\tan 45^\circ = 1$
- $\tan 60^\circ = \sqrt{3}$

What's the range of $\sin\theta$ and $\cos\theta$? $[-1, 1]$

Trigonometric Functions in Different Quadrant

Four Quadrants:



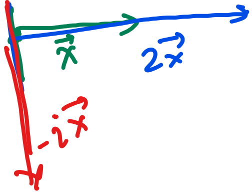
Considering a unit circle, in four different quadrants (x, y) signs are different, Hence the trigonometric ratios are also different in signs.

	Quadrant	$\sin$	$\cos$	$\tan$
+ +	I	+	+	+
− +	II	+	-	-
− −	III	-	-	+
+ −	IV	-	+	-

Table: Signs of $\sin$, $\cos$, and $\tan$ in all four quadrants

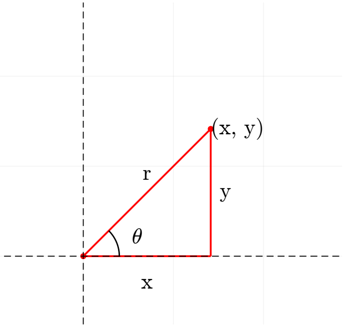
$\cos 135^\circ < 0$

Complex Number



$z = x + iy$ ✓

$i^2 = -1$ $\quad i = \sqrt{-1}$



Two representations of complex numbers:

$z = re^{i\theta}$ Polar

Conjugate: $\bar{z} = re^{-i\theta}$

$\bar{z} = x - iy$

- $r = \sqrt{x^2 + y^2}$
- $\theta = \tan^{-1} \dfrac{y}{x}$

Euler's Identity:

- $\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$
- $e^{-i\theta} = \cos\theta - i\sin\theta$
- $\cos\theta = (e^{i\theta} + e^{-i\theta})/2$ (in terms of exponential)
- $\sin\theta = (e^{i\theta} - e^{-i\theta})/2i$ (in terms of exponential)

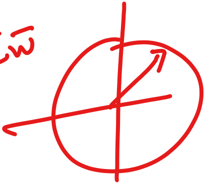
Complex Number

$(3+4i)^{53} = re^{i\theta}$

$z = re^{i\theta}$

Properties of Conjugates

- $\overline{z+w} = \bar{z} + \bar{w}$
- $\overline{zw} = \cancel{\bar{z}\bar{w}}$ $\bar{z}\bar{w}$
- $\overline{z^n} = \bar{z}^n$
- $z\bar{z} = |z|^2$ **Prove it!**



$|z|^2 = x^2+y^2$, magnitude

$z\bar{z} = (x+iy)(x-iy)$

$= x^2 - \cancel{ixy} + \cancel{ixy} - i^2y^2 = x^2+y^2 = |z|^2$

De Moivre's Theorem If $z = r(\cos\theta + i\sin\theta)$ and n is a positive integer, then

$$z^n = [r(\cos\theta + i\sin\theta)^n] = r^n(\cos n\theta + i\sin n\theta)$$

Roots of a complex number $z = 0$ has n distinct roots:

$$w_k = r^{1/n}\left[\cos\left(\frac{\theta+2k\pi}{n}\right) + i\sin\left(\frac{\theta+2k\pi}{n}\right)\right]$$

$re^{i\theta} = 0$

Complex Exponentials

$2! = 1 \times 2$

$3! = 1 \times 2 \times 3$

We also need to give a meaning to the expression e^z when $z = x + iy$ is a complex number. Based on the infinite series using Taylor's series expansion for e^x, we have

$$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots$$

$z = iy$

Can you find out, what should be the value of e^{iy} where y is the real number?

Note: $\cos y = 1 - \frac{y^2}{2!} + \frac{y^4}{4!} - \frac{y^6}{6!} + \cdots$ and $\sin y = y - \frac{y^3}{3!} + \frac{y^5}{5!} - \cdots$

$i^2 = -1$

$i^3 = i^2 \cdot i = -i$

$e^{i\theta} = \cos\theta + i\sin\theta$

$\cos y + i\sin y = e^{iy} = 1 + (iy) + \frac{(iy)^2}{2!} + \frac{(iy)^3}{3!} + \cdots = 1 + iy - \frac{y^2}{2!} - \frac{iy^3}{3!}$

$= \left(1 - \frac{y^2}{2!} + \cdots\right) + \cdots$

Infinite Series

Sum of an Infinite Sequence

An infinite series is the sum of an infinite sequence of numbers:

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

The goal is to understand the meaning of such an infinite sum and develop methods to calculate it.

Partial Sums

The n-th partial sum is:

$$S_n = a_1 + a_2 + a_3 + \cdots + a_n$$

As n gets larger, the partial sums get closer to a limiting value.

Example 1

Consider the ~~series~~: sum

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \cdots \quad - - - \quad \infty$$

The partial sums are:

$$S_1 = 1$$

$$S_2 = 1 + \frac{1}{2} = \frac{3}{2}$$

$$S_3 = 1 + \frac{1}{2} + \frac{1}{4} = \frac{7}{4}$$

$$S_n = 1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}} = \frac{2^n - 1}{2^{n-1}}$$

Geometric Progression

Common Ratio

$$r = \frac{a_n}{a_{n-1}} = \frac{1/4}{1/2}$$

$$r = 1/2$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{1 \cdot (1-(\frac{1}{2})^n)}{1-\frac{1}{2}}$$

First term

Convergence of the Series

$$\frac{a^m}{a^n} = a^{m-n} \qquad \frac{1}{\infty} = 0$$

$$|r| < 1$$

$$S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{2^n - 1}{2^{n-1}} = \lim_{n\to\infty} \left(\frac{2^n}{2^{n-1}} - \frac{1}{2^{n-1}}\right) = \lim_{n\to\infty} \left(2 - \frac{1}{2^{n-1}}\right) \cancel{2} = 2$$

Since the sequence of partial sums converges, the infinite series converges. That is to say:

$$\sum_{n=1}^{\infty} \frac{1}{2^{n-1}} = 2$$

Arithmetic Progression (AP)

An artihmetic series is of the form: $a, (a+d), (a+2d), \cdots (a+(n-1)d)$.
The sum can be written as:

$$S_n = a + (a+d) + (a+2d) + \cdots + (a+(n-1)d) = \frac{n}{2}[2a + (n-1)d]$$

$a =$ The first term, $d =$ common difference.

Geometric Series/Geometric Progression

A geometric series is of the form: $a, ar, ar^2, \cdots ar^n$.
The sum can be written as:

$$S_n = a + ar + ar^2 + ar^3 + \cdots + ar^n + \cdots = \sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=0}^{\infty} ar^n$$

$a =$ The first term, $r =$ common ratio.

Convergence of Geometric Series

If $|r| < 1$, the geometric series converges:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$$

$a = 1$
$r = 1/2$

$$\frac{1}{1-\frac{1}{2}} = 2$$

If $|r| \geq 1$, the series diverges.

Example 2

Consider the series:

$$\sum_{n=0}^{\infty}(-1)^n \frac{5}{4^n} = = a \qquad = \sum\left(-\frac{1}{4}\right)^n \cdot 5$$

This is a geometric series with $a = 5$ and $r = -\frac{1}{4}$. Since $|r| < 1$, the series converges:

$$\sum_{n=1}^{\infty} 5\left(-\frac{1}{4}\right)^{n-1} = \frac{5}{1-\left(-\frac{1}{4}\right)} = 4 \qquad \frac{a}{1-r}$$

$$= \frac{5}{1-\left(-\frac{1}{4}\right)}$$

$$= 4$$

Example 3: Telescoping Series

$S_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots$

$S_n' = \frac{1}{2} + \frac{1}{3} + \cdots$

$S_n - S_n' =$ Telescoping Series

Find the sum of the telescoping series:

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \sum_{n=1}^{\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

The partial sums are:

$$S_n = 1 - \frac{1}{n+1} \to 1 \text{ as } n \to \infty$$

Theorem

If the series $\sum_{n=1}^{\infty} a_n$ converges, then $\lim_{n\to\infty} a_n = 0$.

Divergence Test

The series $\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n \to \infty} a_n$ fails to exist or is different from zero.

Practice Problems

Determine whether the following series converge or diverge. In the case of convergence, give the sum of the series.

1. $\sum_{n=1}^{\infty} \frac{1}{2^n}$ $\rightarrow$ Converging.
2. $\sum_{n=1}^{\infty} \frac{n+1}{2n-3}$
3. $\sum_{n=2}^{\infty} \frac{n^2}{n^2-1}$
4. $\sum_{n=1}^{\infty} \frac{n(n+2)}{(n+3)^2}$
5. $\sum_{n=1}^{\infty} \frac{1+2n}{3^n}$

$$a_n = \frac{1}{2^n}$$

$$n \to \infty \quad \frac{1}{2^\infty} = \frac{1}{\infty} = 0$$

Derivatives

time	m/s	f(t)
0.012	3.1	
0.015	3.3	

$f(t) = e^t$

$h = 0.0001$ = time difference

Rate of change of a quantity with respect to another quantity.

In signals and systems, we are usually interested in the rate of change with respect to time.

$h = 0.015 - 0.012$

at $t = 0.015$

$$f'(t) = \frac{df(t)}{dt} = \lim_{h \to 0} \frac{f(t+h) - f(t)}{h} = \lim_{h \to 0} \frac{f(t) - f(t-h)}{h}$$

In computer implementation, h is Δt. Δt is the difference of timestamps between two consecutive samples of a signal.

$$f'(t)\Big|_{t=0.015} = \frac{f(0.015) - f(0.012)}{0.015 - 0.012} =$$

Derivative Review

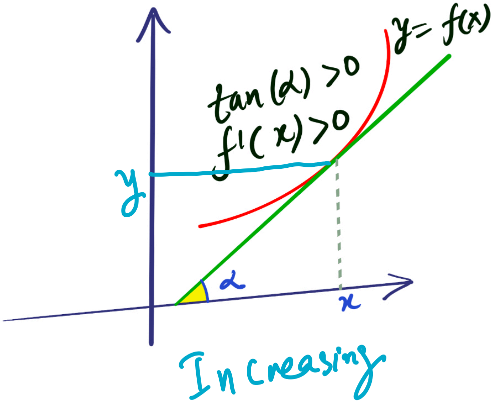
- $f'(x) = \dfrac{df(x)}{dx}$
- $f'(t) =$

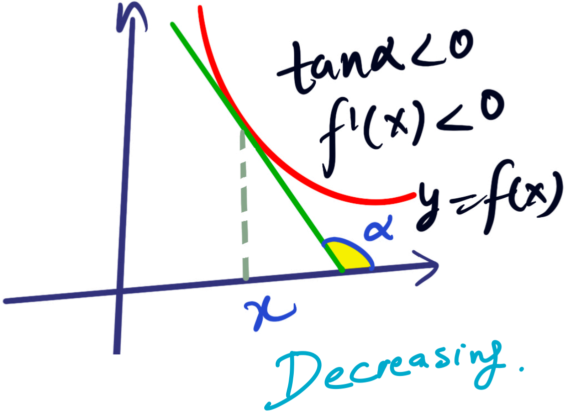
- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\frac{d}{dx}(\ln(x)) = \frac{1}{x}$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(\sin(x)) = \cos(x)$
- $\frac{d}{dx}(\cos(x)) = -\sin(x)$
- $\frac{d}{dx}(\sin^{-1}(x)) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\tan(x)) = \sec^2(x)$
- $\frac{d}{dx}(\cot(x)) = -\csc^2(x)$

- $\frac{d}{dx}(\sec^{-1}(x)) = \frac{1}{x\sqrt{x^2-1}}$
- $\frac{d}{dx}(\sec(x)) = \sec(x)\tan(x)$
- $\frac{d}{dx}(\csc(x)) = -\csc(x)\cot(x)$
- $\frac{d}{dx}(\tan^{-1}(x)) = \frac{1}{x^2+1}$
- $(F \cdot S)' = F \cdot S' + F' \cdot S$
- $\left(\frac{N}{D}\right)' = \frac{D \cdot N' - N \cdot D'}{D^2}$
- $[f(g(x))]' = f'(g(x)) \cdot g'(x)$

Geometrical Interpretation of Derivatives

The derivative $f'(x)$ can be interpreted as the slope of the tangent line at the point (x, y) on the graph of the function $y = f(x)$.

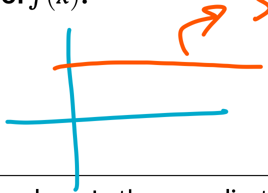




Geometrical Interpretation of Derivatives

We can use the geometric interpretation
to help study the behavior and the graph of $f(x)$.

- $f'(x) > 0$ exactly when $f(x)$ is increasing.
- $f'(x) < 0$ exactly when $f(x)$ is decreasing.
- $f'(x) = 0$ exactly when $f(x)$ has a horizontal tangent.



Saddle or constant value

In many applications, we want to find local maximum and local minimum values. In these applications, it makes sense to look at the locations on a graph where the slope is 0. In other words, we often use the following method to find all locations where the slope of the graph is 0.

- Find the formula for the derivative: $f'(x)$.
- Set this formula equal to 0 and solve for x.

$f'(x) = 0$

Taylor Series

Function approximation:

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

When $a = 0$, the series is called Maclaurin series.

Signal processing examples:

$$\boxed{e^x = 1 + x} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

$$\cos(\omega t) = 1 - \frac{(\omega t)^2}{2!} + \frac{(\omega t)^4}{4!} - \cdots$$

$$\ln(1+t) = t - \frac{t^2}{2} + \frac{t^3}{3} - \cdots \quad (|t| < 1)$$

Taylor Series Applications

$\sin\theta \approx \theta$ $\theta <<< 1$

$\sin\theta \approx \theta - \frac{\theta^3}{6}$ =

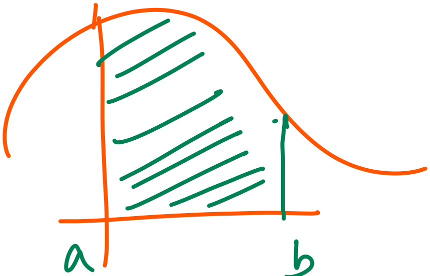
Small-angle approximation:
$\sin\theta \approx \theta - \frac{\theta^3}{6}$ for $\theta \ll 1$
Error at $\theta = 0.1$ rad: $|\text{Actual} - \text{Approx}| < 0.001\%$

Nonlinear system analysis:
$y(t) = e^{x(t)} \approx 1 + x(t) + \frac{x^2(t)}{2}$ ✓
When $x(t) = 0.2\cos(2\pi f t)$,
$y(t) \approx 1 + 0.2\cos(2\pi f t) + 0.02\cos^2(2\pi f t)$
✓

Integration

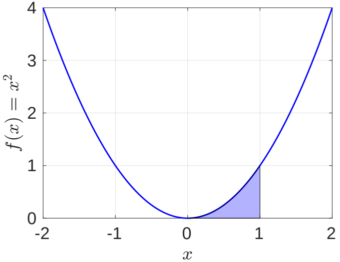
Integration is antiderivative but you can also understand it as an area under the curve. In other words, integration is the summation.

If $\dfrac{d}{dx}F(x) = f(x)$ then $\displaystyle\int f(x)dx = F(x)$.



Geometric Interpretation of Integral

Integral gives the area under a curve



$$\int_0^1 x^2 dx$$

Table of Integrals: Basic Form

- $\int x^n \, dx = \frac{1}{n+1} x^{n+1}$
- $\int \frac{1}{x} \, dx = \ln x$
- $\int u \, dv = uv - \int v \, du$
- $\int u(x) v'(x) \, dx = u(x) v(x) - \int v(x) u'(x) \, dx$

Table of Integrals: More Functions

- $\int \frac{1}{ax+b}\, dx = \frac{1}{a} \ln(ax+b)$
- $\int \frac{1}{(x+a)^2}\, dx = -\frac{1}{x+a}$
- $\int (x+a)^n\, dx = \frac{(x+a)^{n+1}}{n+1}, \quad n \neq -1$
- $\int x(x+a)^n\, dx = \frac{(x+a)^{n+1}(nx+x-a)}{(n+2)(n+1)}$
- $\int \frac{dx}{1+x^2} = \tan^{-1} x$
- $\int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$
- $\int \frac{x\, dx}{a^2+x^2} = \frac{1}{2} \ln\left(a^2+x^2\right)$
- $\int \frac{x^2\, dx}{a^2+x^2} = x - a \tan^{-1}\left(\frac{x}{a}\right)$
- $\int \frac{x^3\, dx}{a^2+x^2} = \frac{1}{2}x^2 - \frac{1}{2}a^2 \ln\left(a^2+x^2\right)$
- $\int e^{ax} = \frac{1}{a} e^{ax}$
- $\int \sin x dx = -\cos x$
- $\int \cos x dx = \sin x$
- $\int \tan x dx = -\ln \cos x$

Differential Equations

Differential equations are simply equations that have derivatives in them.

Example:

- $\dfrac{dy}{dx} + 5y = 0$
- $y'' + 2y' + 10y = 0$
- $y'' + 3y' + 2y = 3\cos t$

The order of a differential equation is the order of the highest derivative in the equation. They are all examples of 'ordinary differential equations'.

Solving Differential Equations

$y' = \frac{dy}{dt}$

$y'(t) = 2y(t) + 3$

y(t) means y is a function of t.

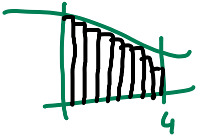
Consider

$$y' = 2y + 3$$

The Fundamental Theorem of Calculus implies $y(t) = \int y'(t)dt$, so we get

$$y(t) = 2\int_0^4 y(t)dt + 3t + C$$

This is not the closed form and doesn't entirely solve to find a solution y. We look at some formula (and their derivation) using the rules of differentiation on how to get a closed-form solution.



Solving Differential Equations

The linear differential equation

$$y' = ay + b$$

with $a \neq 0, b$ has infinitely many solutions

$$y(t) = ce^{at} - \frac{b}{a}$$

Proof discussed in class.

if $a \neq 0$

$y' - ay = b$

we multiply both side by e^{-at}

$y'e^{-at} - aye^{-at} = be^{-at}$

Note that left side is a derivative of ye^{-at}

$\rightarrow \frac{d}{dt}(e^{-at}y) = be^{-at}$

$\Rightarrow e^{-at}y = \int be^{-at}dt = -\frac{b}{a}e^{-at} + c \Rightarrow y(t) = ce^{at} - \frac{b}{a}$

Up Next

- Representation of Signals: Continuous and Discrete
- Continuous Signals
- Using MATLAB